

Rakshak S Barkur

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EDUCATION

BMS Institute of Technology & Management, Bengaluru

BE in Department of Artificial Intelligence and Machine Learning

Relevant Coursework: DSA, AI, DBMS, OS, OOPS with Java, CN, MLOps, MongoDB & Git.

Sep 2023 - May 2027

CGPA - 8.5

TECHNICAL STRENGTHS

Languages : C/C++, Python, Kotlin, HTML, CSS, JavaScript & Java

Tools : VSCode, LaTeX, Excel, Jupyter Notebook & MATLAB

Databases : SQL, MongoDB, Qdrant, FAISS, PostgreSQL & Redis

Libraries : STL (C++), LlamaIndex, Hugging Face Transformers, Flower, Numpy, Scikit-Learn, TensorFlow, Pandas & Matplotlib

Certifications : C++, Python, DSA, Google AI Essential, Google Prompt Engineering, CP & MySQL

EXPERIENCE

Circles - Data Engineer, Intern

Aug'26 - Present

- Developing Python based ELT pipelines using Apache Airflow, Spark, SQL, Iceberg & AWS S3 for scalable data processing.
- Implementing API-driven data integration workflows including XML/SOAP requests, record-level processing, response handling & error management.
- Working with Kubernetes and cloud-native data workflows, focusing on pipeline automation, reliability and data validation.

TestAing - AI/ML, Intern

Mar'26 - Jul'26

- Architected a multi-agent AI framework using LangGraph and FastAPI, with a supervisor node which dynamically routes tasks across four domain-specific specialist agents with parallel execution and a confidence-based decision policies.
- Built real-time agent-frontend state synchronization using AG-UI, SSE, CopilotKit and RFC 6902 JSON Patch, supporting both controlled React components and schema-driven GenUI/A2UI rendering.

Conatus Bharat - Full Stack Development, Intern

Sep'25 - Mar'26

- Developed and modularized a digital education platform using TypeScript, NestJS, Next.js, React, TailwindCSS and PostgreSQL, building REST APIs with JWT/bcrypt authentication.
- Optimized application and PostgreSQL performance through backend refactoring, indexing, constraints, and schema improvements, reducing average latency by 88% (from 25ms to 3ms) and improving throughput.

PROJECTS

Early Detection of Neonatal Sepsis using Federated Learning 📍/Neonatal-Sepsis

Accepted into ICICI'26

- Implemented a privacy-preserving federated learning framework for early neonatal sepsis prediction, enabling secure multi-hospital EHR training using Flower and FedAvg, ensuring HIPAA- and GDPR-compliant collaboration.
- Developed GRU-D and Transformer time-series models on 40 NICU clinical features, improving AUROC from 0.829 → 0.893 and accuracy from 73.9% → 94.7% to enhance early detection reliability.
- Strengthened clinical decision support by increasing precision from 0.187 to 0.712 (~280) and reducing the false-positive rate from 0.2422 to 0.0147, minimising alarm fatigue and unnecessary interventions.

RAG: Agentic Compliance RAG Engine 📍/RAG

- Architected a Python Agentic RAG system featuring a multi-stage reasoning loop that autonomously enhanced query effectiveness, improving answer relevance for over 1,000 unique data sources.
- Engineered a high-precision retrieval engine using Qdrant, implementing Hybrid Search (Dense + BM25) fused via Reciprocal Rank Fusion (RRF) and FlashRank cross-encoder reranking to achieve superior accuracy for both semantic and exact-keyword queries.
- Optimised performance and cost through a tiered LLM strategy, leveraging local Mistral (Ollama) for reasoning and fact-checking while reserving Gemini 2.0 Flash for final synthesis, reducing cloud API consumption by ~80 %.

Folibuddy: AI-Powered Resume to Portfolio Generator 📍/Folibuddy

Live Demo @folibuddy

- Automated conversion of PDF resumes into deployable personal websites, implementing backend in Python, backed with FastAPI and Google Gemini API to orchestrate parsing and site generation.
- Implemented structured LLM outputs for resume parsing and designed an editable input interface enabling users to review & personalise extracted content before publishing the site to a custom domain/URL.

EXTRA CURRICULAR ACTIVITIES

Competitive Programming:

- </> LeetCode : Solved 450+ problems, Peak Rating 1953.
- </> GeeksforGeeks : Solved 700+ problems

Positions of Responsibility: Serving as Chair & VP of the Alterino, a technical Club at BMSIT, leading event organisation and logistics.

Open Source Contributor (GirlScript Summer of Code):

Contributed to open-source projects and secured a global rank of 76 among nearly 50,000 contributors.